

Quantitative Comparisons



Numeric comparisons

- 1 Column A: 7×8 . Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?
 - 2 Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.
 - 3 Column A: 9^2 . Column B: $8^2 + 15$. Compare the two columns.
 - 4 Column A: $(-3)^2$. Column B: -3^2 . Compare the two columns.
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Fraction and decimal comparisons

- 5 Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.
 - 6 Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.
 - 7 Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.
 - 8 Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.
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Percent comparisons

- 9 Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.
 - 10 Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.
 - 11 Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.
 - 12 Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.
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Exponent and root comparisons

- 13 Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.
- 14 Column A: 2^{10} . Column B: 10^3 . Compare the two columns.

15 Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.

16 Column A: 3^4 . Column B: 4^3 . Compare the two columns.

Algebraic comparisons at a fixed value

17 Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

18 Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

19 Given $n = -2$. Column A: n^2 . Column B: n^3 . Compare the two columns.

20 Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.

Comparisons requiring testing values

21 For any real number x : Column A: x . Column B: x^2 . Compare the two columns.

22 For any real number x : Column A: $x + 5$. Column B: $2x$. Compare the two columns.

23 For any real number x : Column A: $|x|$. Column B: x . Compare the two columns.

24 Given only that n is an integer: Column A: n^2 . Column B: 4. Compare the two columns.

25 For any real number x : Column A: $(x - 3)^2$. Column B: $x^2 - 6x + 9$. Compare the two columns.

Geometry comparisons

26 Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

27 Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

28 Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.

29 Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

Ratio and proportion comparisons

- 30** Column A: the value of x , if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.
- 31** A recipe uses flour and sugar in the ratio 3 : 1. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.
- 32** Column A: the sum of the parts in the simplified form of the ratio 18 : 24. Column B: 8. Compare the two columns.
- 33** Two numbers are in the ratio 5 : 3, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

Data and probability comparisons

- 34** A data set is 4, 7, 7, 9, 13. Column A: the mean. Column B: the median. Compare the two columns.
- 35** Column A: the range of the data set 3, 8, 15, 22. Column B: 20. Compare the two columns.
- 36** A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

Word-problem comparisons

- 37** Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.
- 38** Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.
- 39** Column A: the simple interest earned on 2,000 riyals at 5% per year for 3 years. Column B: 350 riyals. Compare the two columns.
- 40** A car travels 240 km using 20 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km per liter. Compare the two columns.

Comparisons with inequalities and ranges

- 41 Given that $5 < x < 9$. Column A: x . Column B: 7. Compare the two columns.
- 42 Given that $x > 10$. Column A: x . Column B: 10. Compare the two columns.
- 43 Given that $0 < x < 1$. Column A: x . Column B: x^2 . Compare the two columns.
- 44 Given that $-5 \leq x \leq -1$. Column A: x^2 . Column B: 10. Compare the two columns.
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More geometry comparisons

- 45 Column A: the volume of a cube with edge 4 cm. Column B: the volume of a rectangular box 3 cm $\times$ 4 cm $\times$ 6 cm. Compare the two columns.
- 46 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the perimeter of a square with side 11 cm. Compare the two columns.
- 47 Column A: the area of a triangle with base 10 and height 8. Column B: the area of a rectangle with length 9 and width 5. Compare the two columns.
- 48 Column A: the sum of the interior angles of a quadrilateral. Column B: three times the sum of the interior angles of a triangle. Compare the two columns.
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MCQ — Numeric comparisons

- 49 Column A: 6×7 . Column B: $5 \times 9 - 3$. Compare the two columns.
- a. The two columns are equal
 - b. Column B is greater
 - c. The relationship cannot be determined from the given information
 - d. Column A is greater

50 Column A: $345 + 678$. Column B: $900 + 100$. Compare the two columns.

- a. Column A is greater
- b. The relationship cannot be determined from the given information
- c. Column B is greater
- d. The two columns are equal

51 Column A: 11^2 . Column B: $10^2 + 23$. Compare the two columns.

- a. Column B is greater
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. The two columns are equal

52 Column A: $(-4)^2$. Column B: -4^2 . Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column B is greater
- d. Column A is greater

MCQ — Fraction and decimal comparisons

53 Column A: $\frac{5}{8}$. Column B: 0.62. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. The two columns are equal
- d. Column A is greater

- 54 Column A: $\frac{9}{16}$. Column B: 0.5625. Compare the two columns.
- a. Column A is greater
 - b. Column B is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information
- 55 Column A: $3\frac{1}{6}$. Column B: $\frac{19}{6}$. Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. Column B is greater
 - c. Column A is greater
 - d. The two columns are equal
- 56 Column A: 0.7. Column B: $\frac{5}{7}$. Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column B is greater
 - d. Column A is greater

MCQ — Percent comparisons

- 57 Column A: 20% of 150. Column B: 25% of 120. Compare the two columns.
- a. The two columns are equal
 - b. Column A is greater
 - c. Column B is greater
 - d. The relationship cannot be determined from the given information

- 58 Column A: 45% of 80. Column B: 40% of 100. Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column A is greater
 - d. Column B is greater
- 59 Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column A is greater
 - d. Column B is greater
- 60 Column A: 18% of 50. Column B: 9% of 90. Compare the two columns.
- a. Column B is greater
 - b. The relationship cannot be determined from the given information
 - c. Column A is greater
 - d. The two columns are equal

MCQ — Exponent and root comparisons

- 61 Column A: $\sqrt{72}$. Column B: 8.5. Compare the two columns.
- a. Column B is greater
 - b. Column A is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information

- 62 Column A: 3^5 . Column B: 5^3 . Compare the two columns.
- a. Column A is greater
 - b. The two columns are equal
 - c. Column B is greater
 - d. The relationship cannot be determined from the given information
- 63 Column A: $\sqrt{0.16}$. Column B: 0.4. Compare the two columns.
- a. Column A is greater
 - b. Column B is greater
 - c. The relationship cannot be determined from the given information
 - d. The two columns are equal
- 64 Column A: 2^6 . Column B: 6^2 . Compare the two columns.
- a. The relationship cannot be determined from the given information
 - b. Column A is greater
 - c. Column B is greater
 - d. The two columns are equal

MCQ — Algebraic comparisons at a fixed value

- 65 Given $x = 6$. Column A: $3x - 4$. Column B: $x + 10$. Compare the two columns.
- a. Column B is greater
 - b. Column A is greater
 - c. The relationship cannot be determined from the given information
 - d. The two columns are equal

66 Given $x = 3$. Column A: $(x + 1)^2$. Column B: $x^2 + 2x + 1$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. Column A is greater
- d. The two columns are equal

67 Given $n = -3$. Column A: n^2 . Column B: n^3 . Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column B is greater
- c. The two columns are equal
- d. Column A is greater

68 Given $x = 2$. Column A: $4x + 1$. Column B: $3(x + 2)$. Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. Column B is greater

MCQ — Comparisons requiring testing values

69 For any real number x : Column A: x^2 . Column B: 0 . Compare the two columns.

- a. The relationship cannot be determined from the given information
- b. The two columns are equal
- c. Column A is greater
- d. Column B is greater

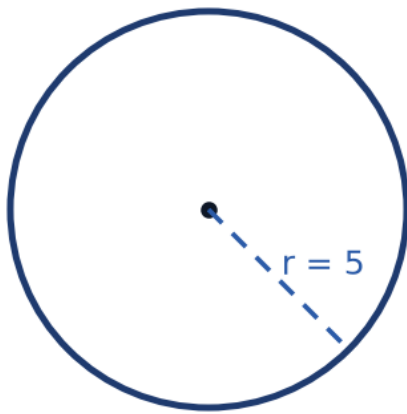
- 70 For any integer n : Column A: n . Column B: $-n$. Compare the two columns.
- a. The two columns are equal
 - b. The relationship cannot be determined from the given information
 - c. Column B is greater
 - d. Column A is greater
- 71 For any real number x with $x \neq 0$: Column A: x^2 . Column B: 0 . Compare the two columns.
- a. Column B is greater
 - b. Column A is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information
- 72 For any real number x : Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.
- a. The two columns are equal
 - b. Column A is greater
 - c. The relationship cannot be determined from the given information
 - d. Column B is greater
- 73 Given only that m is a positive integer: Column A: m . Column B: m^2 . Compare the two columns.
- a. Column B is greater
 - b. Column A is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information

74 For any real number x : Column A: $-x$. Column B: $x - 1$. Compare the two columns.

- a. The two columns are equal
- b. Column A is greater
- c. The relationship cannot be determined from the given information
- d. Column B is greater

MCQ — Geometry comparisons

75 Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.



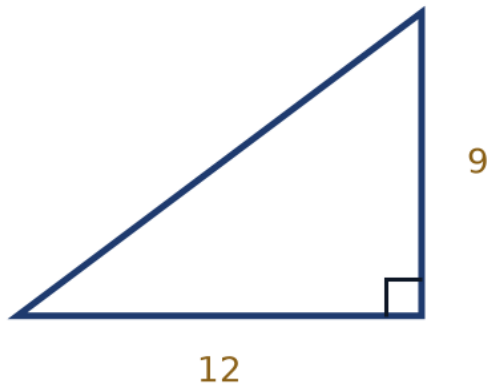
- a. The relationship cannot be determined from the given information
- b. Column A is greater
- c. The two columns are equal
- d. Column B is greater

- 76 Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.



- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column B is greater
 - d. Column A is greater
- 77 Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.
- a. The two columns are equal
 - b. The relationship cannot be determined from the given information
 - c. Column A is greater
 - d. Column B is greater

- 78 Column A: the hypotenuse of the right triangle shown. Column B: 16.



- a. The relationship cannot be determined from the given information
 - b. Column B is greater
 - c. The two columns are equal
 - d. Column A is greater
- 79 Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.
- a. The relationship cannot be determined from the given information
 - b. Column A is greater
 - c. The two columns are equal
 - d. Column B is greater

MCQ — Ratio and proportion comparisons

- 80 Column A: the value of x , if $x : 15 = 2 : 5$. Column B: 7.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column A is greater
 - d. Column B is greater

- 81** Two numbers are in the ratio 4 : 7, and their sum is 99. Column A: the smaller number. Column B: 40.
- a. The two columns are equal
 - b. The relationship cannot be determined from the given information
 - c. Column A is greater
 - d. Column B is greater
- 82** Column A: the sum of the parts in the simplified form of the ratio 24 : 36. Column B: 5.
- a. Column B is greater
 - b. The relationship cannot be determined from the given information
 - c. Column A is greater
 - d. The two columns are equal
- 83** A recipe uses water and rice in the ratio 2 : 1. Column A: the water needed for 300 g of rice. Column B: 550 g.
- a. Column B is greater
 - b. Column A is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information
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MCQ — Data and probability comparisons

- 84** A data set is 5, 8, 8, 10, 14. Column A: the mean. Column B: the median.
- a. The two columns are equal
 - b. Column B is greater
 - c. Column A is greater
 - d. The relationship cannot be determined from the given information

- 85** Column A: the range of the data set 6, 11, 19, 25. Column B: 19.
- a. The relationship cannot be determined from the given information
 - b. Column B is greater
 - c. The two columns are equal
 - d. Column A is greater
- 86** A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.
- a. Column B is greater
 - b. The relationship cannot be determined from the given information
 - c. The two columns are equal
 - d. Column A is greater
- 87** A data set is 2, 4, 4, 4, 6. Column A: the mode. Column B: the mean.
- a. Column A is greater
 - b. The relationship cannot be determined from the given information
 - c. The two columns are equal
 - d. Column B is greater

MCQ — Word-problem comparisons

- 88** Column A: the total cost of 5 items at 20 riyals each, after a 15% discount on the total. Column B: 90.
- a. The two columns are equal
 - b. Column A is greater
 - c. The relationship cannot be determined from the given information
 - d. Column B is greater

- 89 Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.
- a. Column A is greater
 - b. The two columns are equal
 - c. Column B is greater
 - d. The relationship cannot be determined from the given information
- 90 Column A: the simple interest on 3,000 riyals at 4% per year for 2 years. Column B: 250.
- a. The two columns are equal
 - b. The relationship cannot be determined from the given information
 - c. Column A is greater
 - d. Column B is greater
- 91 A car travels 300 km using 25 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km.
- a. The two columns are equal
 - b. The relationship cannot be determined from the given information
 - c. Column B is greater
 - d. Column A is greater

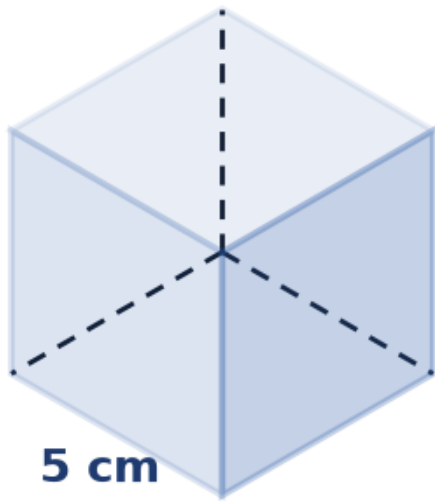
MCQ — Comparisons with inequalities and ranges

- 92 Given that $3 < x < 8$. Column A: x . Column B: 5.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column A is greater
 - d. Column B is greater

- 93 Given that $x < -4$. Column A: x . Column B: -4 .
- a. Column A is greater
 - b. Column B is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information
- 94 Given that $-3 \leq x \leq 2$. Column A: x^2 . Column B: 9.
- a. Column B is greater
 - b. Column A is greater
 - c. The two columns are equal
 - d. The relationship cannot be determined from the given information
- 95 Given that $x \geq 7$. Column A: x^2 . Column B: 49.
- a. The relationship cannot be determined from the given information
 - b. The two columns are equal
 - c. Column B is greater
 - d. Column A is greater

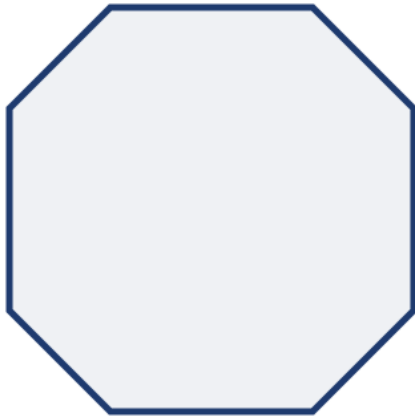
MCQ — More geometry comparisons

- 96 Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.



- a. The relationship cannot be determined from the given information
 - b. Column A is greater
 - c. Column B is greater
 - d. The two columns are equal
- 97 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the area of a square with side 11 cm.
- a. The two columns are equal
 - b. Column A is greater
 - c. Column B is greater
 - d. The relationship cannot be determined from the given information

- 98 Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.



- a. Column B is greater
- b. The two columns are equal
- c. Column A is greater
- d. The relationship cannot be determined from the given information